DEEPESH SURESH

MECHATRONICS ENGINEERING | ROBOTICS | EMBEDDED SYSTEMS | INDUSTRIAL AUTOMATION

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PROFESSIONAL PROFILE

Engineering Sciences graduate specializing in robotics, with hands-on experience designing and integrating underwater robots, quadcopters, gesture-driven manipulators and ESP32-based control systems. Combines mechanical CAD, embedded electronics, sensor integration, Python/C++, controller configuration and prototype testing. Industrial internships at Lynxeo and Emerson strengthened engineering validation, technical documentation and cross-functional problem-solving. Seeking mechatronics, robotics or automation engineering roles.

EDUCATION

École Centrale de Nantes | B.Sc. Engineering Sciences (Robotics Orientation) Sep 2023 - Sep 2026

Nantes, France | Cumulative average: 18.19/20 | 180 validated ECTS

Relevant coursework: Automatic Control, Dynamics, Electrical Engineering, Signal Processing, Numerical Analysis, Scientific Programming, FEM.

IIT Patna | Online Certificate in Artificial Intelligence & Machine Learning Feb - Jul 2026

CGPA: 8.68/10 | Certificate of Excellence in Evaluation

British School of Paris | A Levels: Mathematics (A), Physics (A)

ENGINEERING EXPERIENCE

Lynxeo | Technical Project Intern, Energy Transition & Industrial Cable Systems Jun - Aug 2026

Puteaux, France

- Consolidated cable type-test data, datasheets and technical references, improving consistency and traceability of engineering information.
- Reviewed technical requirements, project documentation and deliverables with specialists using structured verification checks.
- Proposed a simplified finite-element calculation approach and communicated methods and findings to a factory team.

Emerson | Mechanical Engineering Intern, Industrial Automation Jun - Jul 2025

Chennai, India

- Supported industrial automation engineering workflows through technical review, documentation and systematic validation.
- Maintained engineering revision-control records, tracked technical changes and documented handovers for traceable deliverables.

SELECTED MECHATRONICS PROJECTS

Custom 6-DOF Vector-Thrust Underwater ROV | ESP32-S3, Raspberry Pi 4B, Embedded Control

- Co-developed a custom underwater ROV using four rotating vector thrusters (0°-90°) for six-axis manoeuvrability; integrated mechanical design, electronics and waterproof enclosure concepts.
- Integrated embedded control, 4-in-1 bidirectional ESC, Li-ion power, depth hold / stabilization modes, remote control, Ethernet video streaming and lighting.

Pixhawk-Based 4-DOF Underwater ROV | Pixhawk 2.4.8, ArduSub, Sensor Integration

- Designed and built a four-thruster ROV using a custom, modular Simple ROV-4-inspired frame; tuned near-neutral buoyancy for stable, efficient operation.
- Configured ArduSub on Pixhawk 2.4.8, added water-leak detection and reported approximately 45-60 minutes of battery endurance depending on operating load.

Hand-Gesture Controlled Pick-and-Place Robot | Arduino Nano, C++, Python, Servos

- Built a glove-controlled robotic arm using a flex sensor, Arduino Nano and SG90 servos; implemented sensor filtering, gesture-to-joint calibration and PWM actuation.
- Integrated mechanical hardware, wiring and Python sequence logic; demonstrated pick-and-place motions with reported ~120-150 ms response latency.

Smart IoT Water Monitoring & Control System | ESP32, Altium, Java, Android, Wi-Fi

- Built an ESP32-based water system monitor with remote Android app controls for water level, flow rate, volume and motor operation via Wi-Fi.
- Designed a PCB layout in Altium Designer and assembled a physical controller with push-button inputs, sensor connections and a 16x2 LCD; tested the water-flow setup.

Custom F450 Quadcopter | APM 2.8, Mission Planner, Brushless Motors

- Assembled an F450-frame quadcopter integrating an APM 2.8 flight controller, 1000KV brushless motors, ESC wiring and FlySky remote-control interface.
- Configured and tuned flight-control parameters in Mission Planner; planned upgrades include GPS waypoint navigation, FPV, LoRa telemetry and custom firmware.

5-DOF Robotic Arm - Mechanical Design | SolidWorks, MG995/MG90S Servos

- Designed a modular five-degree-of-freedom robotic arm, with higher-torque MG995 servos at main joints and MG90S micro servos at the wrist and gripper.
- Modelled links, joints, bases and end-effector geometry; constrained and motion-tested the CAD assembly for manipulation tasks (design/simulation stage).

AI Train Obstacle Detection & Safety Prototype | Computer Vision, AI, Spatial Analysis

- Developed a capstone computer-vision application using rail-track region-of-interest segmentation and multi-class detection of animals, vehicles and debris.
- Mapped obstacle position / proximity to three alert states: Track Clear, Warning and Danger; designed as an assistance prototype, not a certified safety product.

Computer-Vision Gesture Interfaces | Python, OpenCV, MediaPipe, Real-Time HCI

- Created real-time hand-tracking interfaces that translate camera-detected motion into digital commands for touchless game control and navigation.
- Extended gesture-based interaction to full-body pose recognition for hands-free Subway Surfers actions and a browser-based spaceship game.

Mechanical Kinematics & CAD Mechanisms | Autodesk Fusion 360, Assembly Motion

- Modelled and animated Geneva, Scotch Yoke, Trammel of Archimedes and wheel mechanisms to study indexing, reciprocating / elliptical motion and component interaction.
- Applied assembly joints, dimensional constraints and motion limits; tested the mechanism behaviour in CAD animation environments.

Smartphone Motion-Controlled Interactive Game | Python, IMU Sensors, WebSockets

- Developed a desktop fruit-slicing game controlled by live smartphone gyroscope / accelerometer data, replacing a conventional input device with hand motion.
- Streamed sensor data to a laptop over local WebSockets / sockets and mapped motion angles to screen coordinates in real time.

NeuroPlay Algorithmic Game Arcade | JavaScript, Search Algorithms, Game Theory

- Developed a browser-based arcade with independent game engines using vanilla HTML / CSS / JavaScript and deterministic game-state updates.
- Implemented Minimax / Alpha-Beta Pruning for Connect 4, misere strategy for Pop It and risk-aware move selection for Dots & Boxes.

TECHNICAL SKILLS

Robotics & embedded: ESP32 / ESP32-S3, Arduino Nano, Raspberry Pi, Pixhawk / APM, ArduSub, Mission Planner, ROS, PWM / ESC, sensor integration, controller tuning.

CAD & simulation: SolidWorks, Fusion 360, CATIA V5, AutoCAD, FEM, MATLAB, mechanism design, assembly constraints and engineering validation.

Programming & HCI: Python, C++, Java, OpenCV, MediaPipe, SQL, JavaScript, Git, Linux, Android Studio, real-time sensor processing.

Languages: Tamil (native), English (fluent), French (B1), German (learning).